ds4eeb Final Project

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2026-03-09

In this script we are going to analyzing data from the Kroeker Labs annual surveys of rocky subtidal reef communites near Sitka, Alaska. Over the past few years, we have seen the transition of these reefs from kelp forests to urchin barrens, and the loss of abalone populations. So the motivating questions here are:

Quantitatively, can we see these trends in our data?

-And-

What is the relationship between urchin and abalone populations?

Our target audience is the general public and scientific community as we plan to share these results (after some refinement) with the community of Sitka, the scientific community, and, perhaps, resource managers.

Note: we are only using data from summer surveys, as winter data may obscure our results due to the seasonal fluctuations in algal and fauna communities. So, let's start loading some libraries and read in the data:

Loading libraries

```
library(tidyverse)
library(janitor)
library(here)
library(readr)
library(purrr)
library(scales)
library(lubridate)
```

```r
library(RColorBrewer)
library(brms) # for statistics
library(dagitty) # for loading DAGs
library(patchwork)
library(tidybayes)
```

Reading and plotting the data

```r
swath_density <- read_csv(here("data/swath_density.csv")) %>%
  mutate(
    transect = as.character(transect),
    transect = as.numeric(case_when(
      transect == "upcoast"   ~ "1",
      transect == "downcoast" ~ "2",
      TRUE ~ transect
    )),
    zone = recode(zone, "perm" = "deep")
  ) %>%
  filter(zone != "mid", season == "summer") %>%  # filter out odd zone code, and only include
  select(-season) #remove season column as only using summer data

swath_size_freq <- read_csv(here("data/swath_size_freq.csv")) %>%
  mutate(
    transect = as.character(transect),
    transect = as.numeric(case_when(
      transect == "upcoast"   ~ "1",
      transect == "downcoast" ~ "2",
      TRUE ~ transect
    )),
    zone = recode(zone, "perm" = "deep")
  ) %>%
  filter(zone != "mid", segment != "all", season == "summer") %>%  # filter out odd zone and
  select(-season)

# make key of all real sampling events to fill zeros later
sampling_key <- swath_size_freq %>%
  distinct(year, site, zone, transect, segment)
```

Abalone biomass data

Using the equation W = 0.0000578L^3.2 from Zhang et al 2007 to convert shell length (L in mm) into body weight (W in g). The two parameter values were the means estimated by Breen & Adkins (1982).

Citation: Zhang, Z., Campbell, A., & Lessard, J. (2007). Modeling northern abalone, Haliotis kamtschatkana, population stock and recruitment in British Columbia. Journal of Shellfish Research, 26(4), 1099-1107.

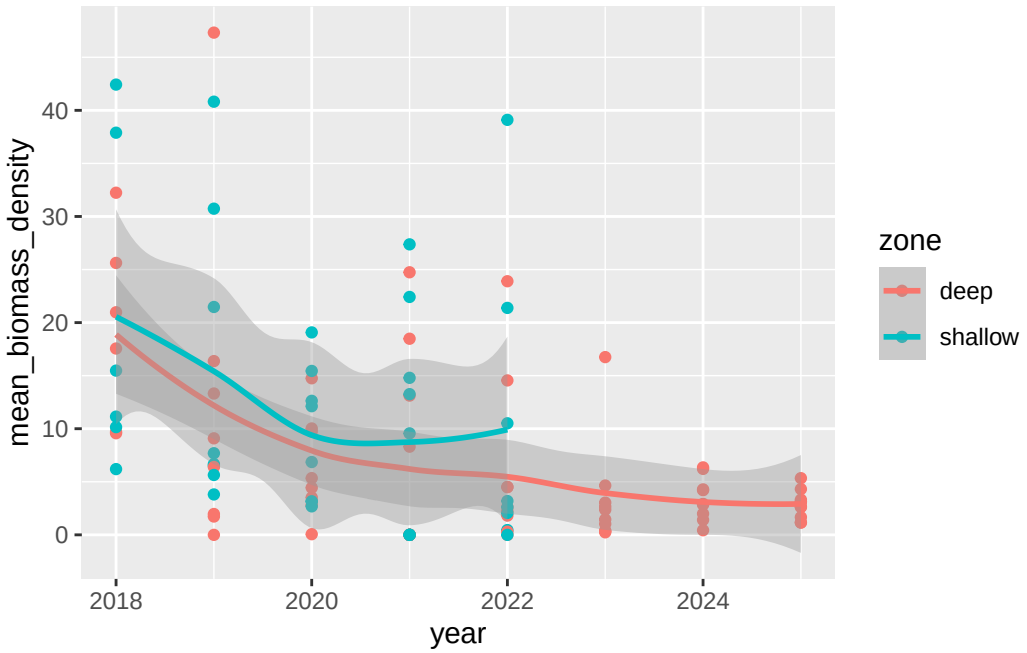
Pull in swath size frequency data, filter for abalone, sum at the segment level, calculate average biomass density (g/m^2) at the transect level for plotting.

```
ab_biodensity_segment <- swath_size_freq %>%
  filter(classcode == "halkam") %>% #filter for abalone
  mutate(biomass_g = 0.0000578 * size_or_stipe_count^3.2) %>%
  group_by(year, site, zone, transect, segment) %>%
  summarise(
    biomass_density = sum(biomass_g, na.rm = TRUE) / 20,
    .groups = "drop"
  ) %>%
  right_join(sampling_key, by = c("year", "site", "zone", "transect", "segment")) %>%
  mutate(biomass_density = replace_na(biomass_density, 0))

ab_biodensity_transect <- ab_biodensity_segment%>%
  group_by(year, site, zone, transect) %>%
  summarise(mean_biomass_density = mean(biomass_density, na.rm = TRUE), .groups = "drop")
```

Plotting Abalone

```
ggplot(ab_biodensity_transect,
       aes(x = year, y = mean_biomass_density, color = zone)) +
  geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Macro biomass density

Constants for macro from Lauren Bell's paper: macro_wet_slope <- 734.9 macro_dry_slope <- 0.103 survey_area_m2 <- 30 * 2 # 60m² segment_area_m2 <- survey_area_m2 / 3 # 20m²

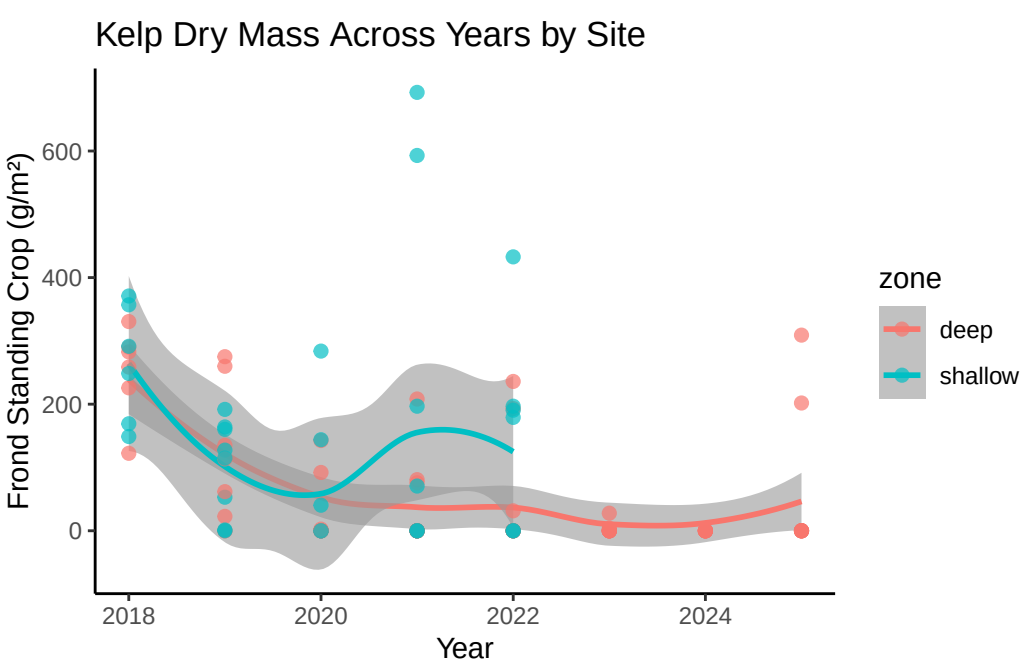
```
macro_wet_slope <- 734.9
macro_dry_slope <- 0.103
#transect_area_m2 <- 20 * 3 # 60m²
#segment_area_m2 <- 10 * 2 # 20m²

macro_biodensity_segment <- swath_size_freq %>%
  filter(classcode == "macpyrad") %>%
  mutate(dry_mass_g_plant = size_or_stipe_count * macro_wet_slope * macro_dry_slope) %>%
  group_by(year, site, zone, transect, segment) %>%
  summarise(
    biomass_density = (sum(dry_mass_g_plant, na.rm = TRUE) / 20),
    .groups = "drop"
  ) %>%
  right_join(sampling_key, by = c("year", "site", "zone", "transect", "segment")) %>%
  mutate(biomass_density = replace_na(biomass_density, 0))
```

```r
macro_biodensity_transect <- macro_biodensity_segment %>%
  group_by(year, site, zone, transect) %>%
  summarise(mean_biomass_density = mean(biomass_density, na.rm = TRUE), .groups = "drop")
```

Plotting Macrocystis

```r
ggplot(macro_biodensity_transect,
       aes(x = year, y = mean_biomass_density, color = zone)) +
  geom_smooth(alpha = 0.6) +
  geom_point(size = 2, alpha = 0.7) +
  #facet_wrap(~ site, scales = "free_y") +
  labs(
    x = "Year",
    y = "Frond Standing Crop (g/m²)",
    title = "Kelp Dry Mass Across Years by Site") +
  theme_classic()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Urchin Data

converting test diameter to biomass

Conversion factors from Lochead, J., W. Hajas, and Don Leus. Calculation of mean abundance in the Red Urchin Analysis Program and Green Urchin Analysis Program. Fisheries and Oceans Canada, Science Branch, Pacific Region, Pacific Biological Station, 2015.

Reds (and using for purps): W = 0.0010169 × T^2.7787 Greens: W = 11.2181 × (T/30)^2.754085

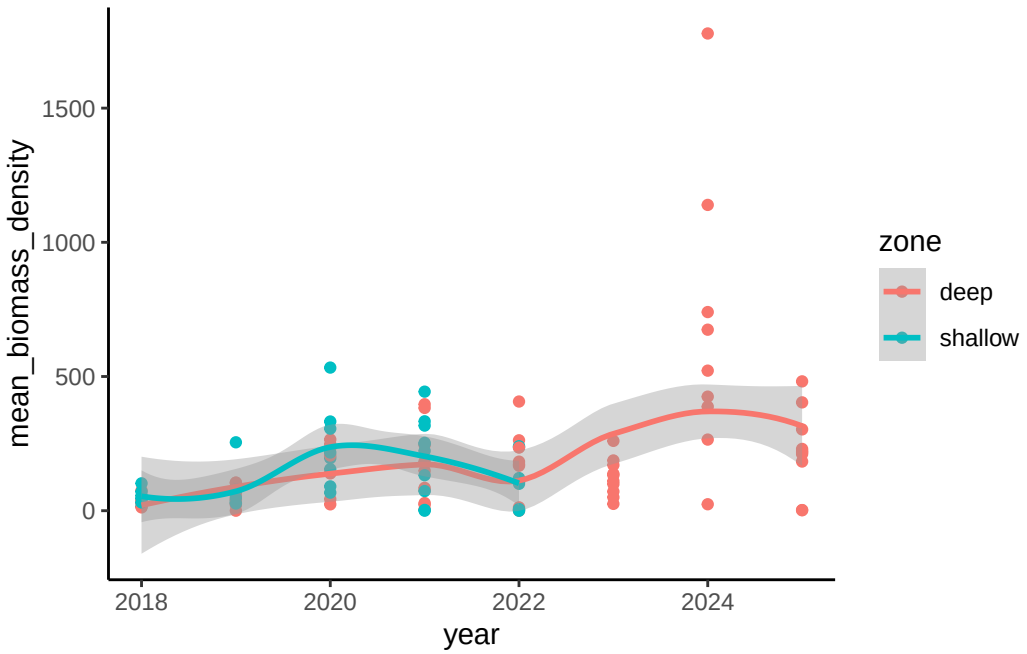
```
custom_colors <- c("Green Urchin" = "#90ee90",
                   "Red Urchin" = "red")


urch_biodensity_segment_species <- swath_size_freq %>%
  filter(classcode %in% c("strdro","strfraad")) %>%
  filter(size_or_stipe_count < 150) %>% #removing outliers
  mutate(
    common_name = case_when(classcode == "strdro" ~ "Green Urchin",
                            classcode == "strfraad" ~ "Red Urchin"),
    site = case_when(site == "Breast" ~ "BREAST", #cleaning
                     site == "Harris" ~ "HARRIS",
                     .default = site),
    biomass_density = case_when(
      classcode == "strdro" ~ 11.2181 * (size_or_stipe_count / 30)^2.754085,
      classcode %in% c("strfraad") ~ 0.0010169 * size_or_stipe_count^2.7787,
    TRUE ~ NA_real_
  )) %>%
  # summing at the segment level by species and divide by area sampled
  group_by(year, site, zone, transect, segment, common_name) %>%
  summarise(biomass_density= sum(biomass_density, na.rm = TRUE) / first(area_samp),
            .groups = "drop") %>%
  right_join(sampling_key, by = c("year", "site", "zone", "transect", "segment")) %>%
  mutate(biomass_density = replace_na(biomass_density, 0))

#1: Sum both species within each segment
urch_biodensity_segment <- urch_biodensity_segment_species %>%
  group_by(year, site, zone, transect, segment) %>%
  summarise(biomass_density = sum(biomass_density, na.rm = TRUE), .groups = "drop")

#2: Average segments up to transect level
urch_biodensity_transect <- urch_biodensity_segment %>%
```

```
  group_by(year, site, zone, transect) %>%
  summarise(mean_biomass_density = mean(biomass_density, na.rm = TRUE), .groups = "drop")
```

Plotting Urchins

```
ggplot(urch_biodensity_transect, aes(x = year, y = mean_biomass_density, color = zone)) +
  geom_point() +
  geom_smooth() +
  theme_classic()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Understory Algae Data

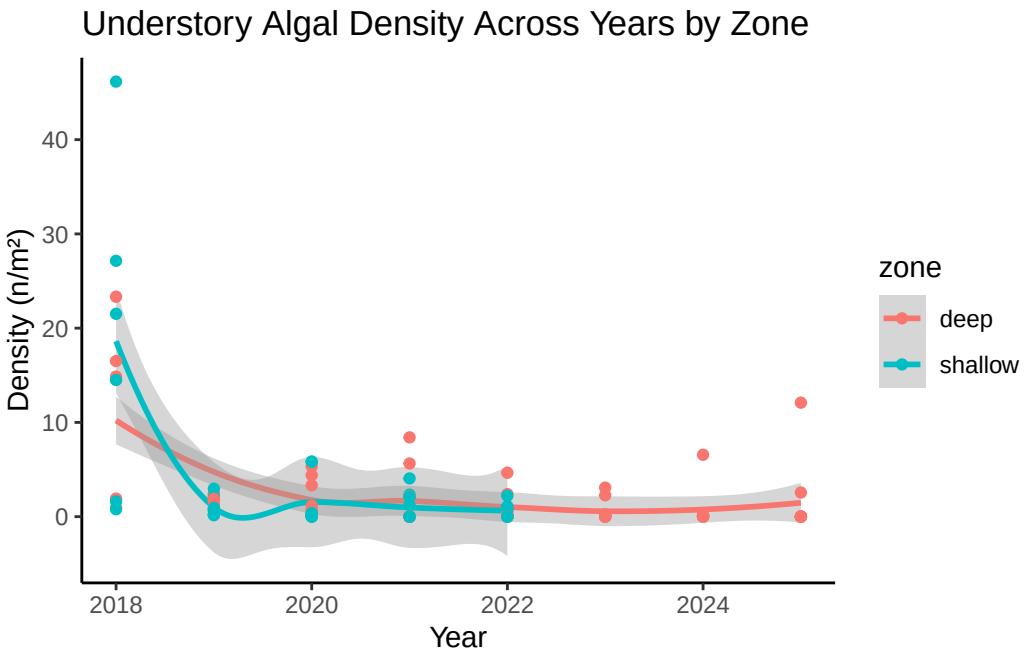
Understory algae is simply density (n/m^2), summed across subcanopy taxa as we do not have biomass conversions for most species.

```r
under_density_segment <- swath_density %>% #reading from density df as we are not converting
  filter(class == "algae", !classcode %in% c("macpyrad", "macpyrjv", "unkalg")) %>% #Remove l
  group_by(year, site, zone, transect, segment) %>%
  summarise(
    under_density = (sum(density, na.rm = TRUE)),
    .groups = "drop"
  ) %>%
  right_join(sampling_key, by = c("year", "site", "zone", "transect", "segment")) %>%
  mutate(under_density = replace_na(under_density, 0))

under_density_transect <- under_density_segment %>%
group_by(year, site, zone, transect) %>%
  summarise(mean_under_density = mean(under_density, na.rm = TRUE), .groups = "drop")
```

Plotting Understory

```r
ggplot(under_density_transect,
       aes(x = year, y = mean_under_density, color = zone)) +
  geom_smooth() +
  geom_point() +
  labs(
    x = "Year",
    y = "Density (n/m²)",
    title = "Understory Algal Density Across Years by Zone") +
  theme_classic()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Pycno Data

pycno biomass conversion = exp(-3.9989)*(length.cm^3.133)

Lee, L. C., J. C. Watson, R. Trebilco, and A. K. Salomon. 2016. Indirect effects and prey behavior mediate interactions between an endangered prey and recovering predator. Ecosphere 7(12):e01604. 10.1002/ecs2.1604

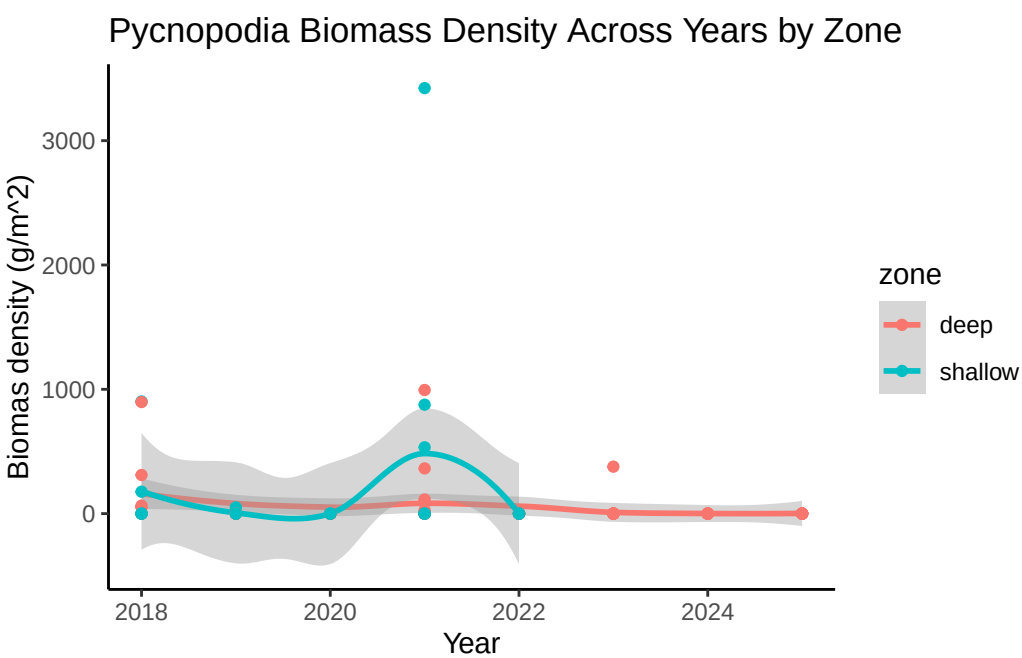
```
#Pycno biomass calculations

pyc_biodensity_segment <- swath_size_freq %>%
  filter(classcode == "pychel") %>%
  mutate(biomass_g = exp(-3.9989) * (size_or_stipe_count^3.133)) %>%
  group_by(year, site, zone, transect, segment) %>%
  summarise(
    biomass_density = sum(biomass_g, na.rm = TRUE) / 20,
    .groups = "drop"
  ) %>%
  mutate(biomass_density = replace_na(biomass_density, 0))%>%
  right_join(sampling_key, by = c("year", "site", "zone", "transect", "segment")) %>%
  mutate(biomass_density = replace_na(biomass_density, 0))

pyc_biodensity_transect <- pyc_biodensity_segment %>%
```

```
  group_by(year, site, zone, transect) %>%
  summarise(mean_biomass_density = mean(biomass_density, na.rm = TRUE), .groups = "drop")
```

Plotting Pycno Data

```
ggplot(pyc_biodensity_transect,
       aes(x = year, y = mean_biomass_density, color = zone)) +
  geom_smooth() +
  geom_point() +
  labs(
    x = "Year",
    y = "Biomas density (g/m^2)",
    title = "Pycnopodia Biomass Density Across Years by Zone") +
  theme_classic()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Putting it all together

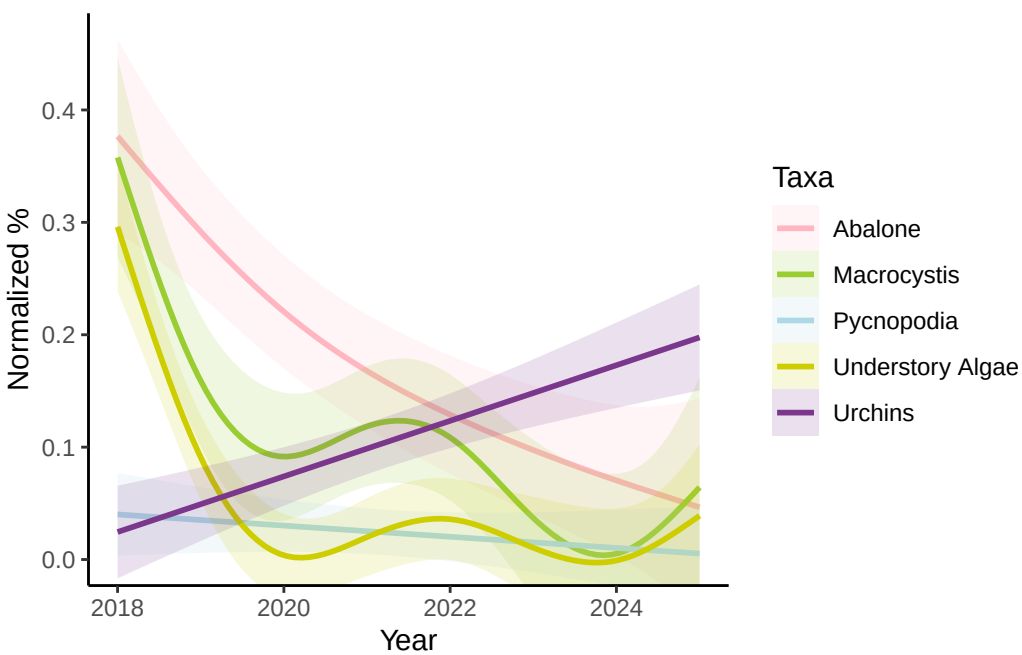
Combining all data

```
#at segment level for model
combined_segments <- bind_rows(
  ab_biodensity_segment %>%
    mutate(taxa = "Abalone"),
  macro_biodensity_segment %>%
    mutate(taxa = "Macrocystis"),
  pyc_biodensity_segment %>%
    mutate(taxa = "Pycnopodia"),
  under_density_segment %>%
    rename(biomass_density = under_density) %>%
  mutate(taxa = "Understory"),
  urch_biodensity_segment %>%
    mutate(taxa = "Urchins"))

#At transect level for plotting
combined_transects <- bind_rows(
  ab_biodensity_transect %>%
    mutate(taxa = "Abalone"),
  macro_biodensity_transect %>%
    mutate(taxa = "Macrocystis"),
  urch_biodensity_transect %>%
    mutate(taxa = "Urchins"),
  under_density_transect %>%
    rename(mean_biomass_density = mean_under_density) %>%
    mutate(taxa = "Understory Algae"),
  pyc_biodensity_transect %>%
    mutate(taxa = "Pycnopodia")
)
```

```
#Normalizing data, max = 1, 0 = 0
combined_normal0 <- combined_transects %>%
  group_by(taxa) %>%
  mutate(mean_scaled = mean_biomass_density / max(mean_biomass_density, na.rm = TRUE)) %>%
  ungroup()
```

Plotting taxa over time

```
taxa_colors <- c(
  "Urchins"       = "mediumorchid4",
  "Abalone"          = "#FFB6C1",
  "Understory Algae" = "yellow3",
  "Macrocystis"      = "olivedrab3",
  "Pycnopodia"  = "lightblue"
)

ggplot(combined_normal0, aes(x = year, y = mean_scaled, color = taxa, fill = taxa)) +
  geom_smooth(method = "gam", formula = y ~ s(x, k = 5), se = TRUE, alpha = 0.15) +
  scale_color_manual(values = taxa_colors) +
  scale_fill_manual(values = taxa_colors) +
  coord_cartesian(ylim = c(0, NA)) +
  labs(x = "Year", y = "Normalized %", color = "Taxa", fill = "Taxa") +
  theme_classic()
```



Plotting urchins against other taxa

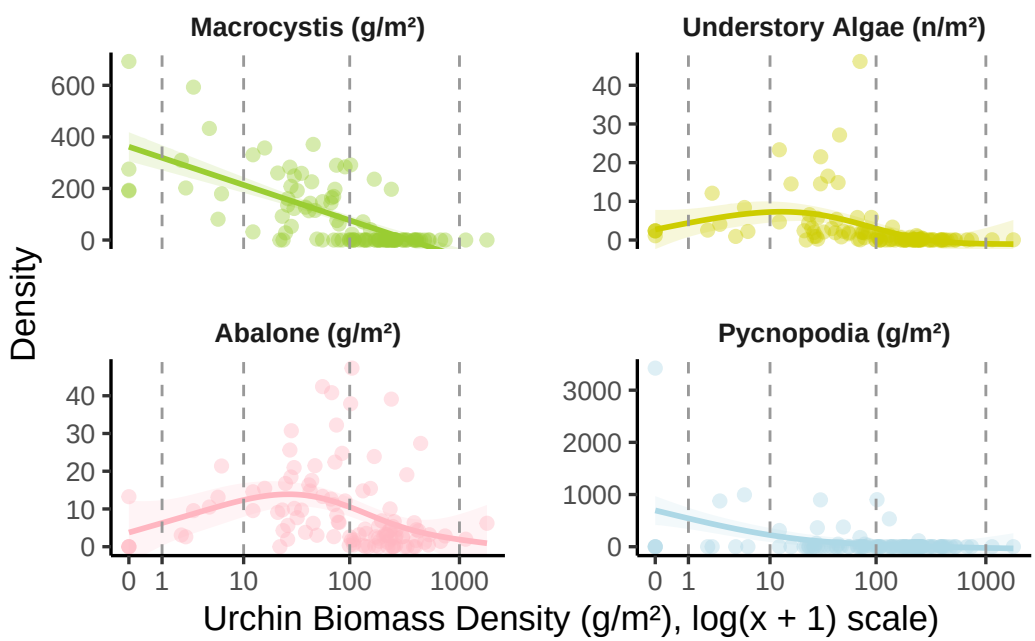
Reformatting data for model

```
# pivot to wide so each taxa becomes its own column,
# allowing us to use Urchins as the x variable in the plot
combined_wide_plot <- combined_transects %>%
  group_by(year, site, zone, transect, taxa) %>%
  summarise(mean_biomass_density = mean(mean_biomass_density, na.rm = TRUE), .groups = "drop
  pivot_wider(names_from = taxa,
              values_from = mean_biomass_density)

# pivot back to long but exclude Urchins - it stays as its own column (the x axis)
combined_long_plot <- combined_wide_plot %>%
  pivot_longer(cols = c("Abalone", "Macrocystis", "Understory Algae", "Pycnopodia"),
               names_to = "taxa",
               values_to = "mean_biomass_density")
```

Plotting Urchins V Other Taxa

```
ggplot(combined_long_plot, aes(x = Urchins, y = mean_biomass_density, color = taxa, fill = ta
  geom_point(alpha = 0.4, size = 2) +
  geom_smooth(method = "gam", formula = y ~ s(x, k = 5), se = TRUE, alpha = 0.15) +
  scale_x_continuous(
    trans = "log1p",
    breaks = c(0, 1, 10, 100, 1000),
    labels = c("0", "1", "10", "100", "1000")
  ) +
  geom_vline(xintercept = c(1, 10, 100, 1000), linetype = "dashed", color = "grey60") +
  facet_wrap(~ factor(taxa, levels = c("Macrocystis", "Understory Algae", "Abalone", "Pycnopo
             scales = "free_y",
             labeller = as_labeller(c(
               "Macrocystis"      = "Macrocystis (g/m²)",
               "Understory Algae" = "Understory Algae (n/m²)",
               "Abalone"          = "Abalone (g/m²)",
               "Pycnopodia"       = "Pycnopodia (g/m²)"
             ))) +
  scale_color_manual(values = taxa_colors) +
  scale_fill_manual(values = taxa_colors) +
```

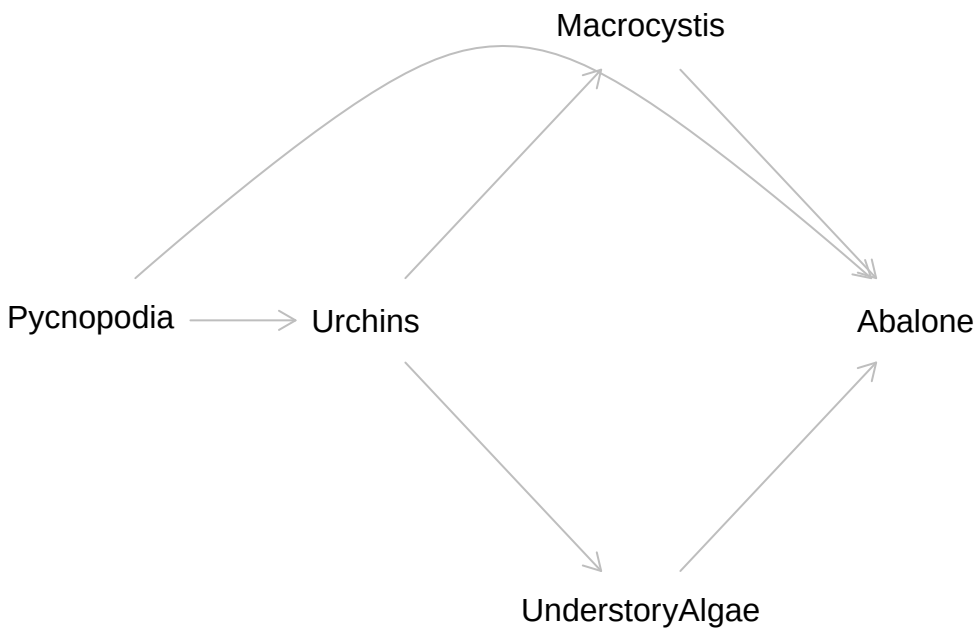
```r
  coord_cartesian(ylim = c(0, NA)) +
  labs(x = "Urchin Biomass Density (g/m²), log(x + 1) scale)",
       y = "Density") +
  theme_classic(base_size = 13) +
  theme(
    legend.position = "none",
    strip.background = element_blank(),
    strip.text = element_text(face = "bold"),
    panel.spacing = unit(1.5, "lines")
  )
```



Our DAG

```
DAG <- dagitty('dag {
bb="0,0,1,1"
Pycnopodia [adjusted,pos="0.100,0.500"]
Urchins [exposure,pos="0.300,0.500"]
Macrocystis [pos="0.500,0.350"]
UnderstoryAlgae [pos="0.500,0.650"]
Abalone [outcome,pos="0.700,0.500"]
```

```
Macrocystis -> Abalone
Pycnopodia -> Abalone [pos="0.400,0.300"]
Pycnopodia -> Urchins
UnderstoryAlgae -> Abalone
Urchins -> Macrocystis
Urchins -> UnderstoryAlgae
}')
plot(DAG)
```



The BRM Model

```
# ------------------------------------------------------------
# 1. Build modeling dataframe
# ------------------------------------------------------------

# Pivot wide so each taxa is its own column
combined_wide_model <- combined_segments %>%
  pivot_wider(names_from = taxa,
              values_from = biomass_density) %>%
  clean_names() %>%  # janitor::clean_names() to handle spaces/caps
```

```r
  mutate( site = factor(site),
    zone = factor(zone),
    transect = factor(transect),
    segment = factor(segment),
    year = factor(year),
    macrocystis = log1p(macrocystis),
    understory = log1p(understory),
    pycnopodia = log1p(pycnopodia),
    urchins = log1p(urchins)
  )

# Check structure
head(combined_wide_model)
```

```
# A tibble: 6 x 10
  year  site   zone  transect segment abalone macrocystis pycnopodia understory
  <fct> <fct>  <fct> <fct>    <fct>     <dbl>       <dbl>      <dbl>      <dbl>
1 2018  breast deep  1        0-10m      29.9        6.10       0         0.560
2 2018  breast deep  1        10-20m     17.1        5.74       0         0.438
3 2018  breast deep  1        20-30m     29.8        4.56       5.27      0.936
4 2018  breast deep  2        0-10m      42.7        5.97       0         0.956
5 2018  breast deep  2        10-20m     30.2        5.49       0         0.896
6 2018  breast deep  2        20-30m     23.8        5.48       0         1.31
# i 1 more variable: urchins <dbl>
```

```r
# ------------------------------------------------------------
# 2. Define model formulas (trophic cascade DAG)
# ------------------------------------------------------------

# Pycnopodia → Urchins
f_urchins <- bf(
  urchins ~ pycnopodia +
    (1 | year) +
    (1 | site/zone/transect/segment),
  family = hurdle_gamma()
)

# Urchins → Macrocystis
f_macrocystis <- bf(
  macrocystis ~ urchins +
    (1 | year) +
```

```r
    (1 | site/zone/transect/segment),
  family = hurdle_gamma()
)

# Urchins → Understory algae
f_understory <- bf(
  understory ~ urchins +
    (1 | year) +
    (1 | site/zone/transect/segment),
  family = hurdle_gamma()
)

# Kelp + predators → Abalone
f_abalone <- bf(
  abalone ~ macrocystis + understory + pycnopodia +
    (1 | year) +
    (1 | site/zone/transect/segment),
  family = hurdle_gamma()
)

# ------------------------------------------------------------
# 3. Priors
# ------------------------------------------------------------

priors <- c(
  prior(normal(0,1), class = "b"),
  prior(normal(0,2), class = "Intercept"),

  prior(exponential(1), class = "shape", resp = "urchins"),
  prior(exponential(1), class = "shape", resp = "macrocystis"),
  prior(exponential(1), class = "shape", resp = "understory"),
  prior(exponential(1), class = "shape", resp = "abalone")
)

# ------------------------------------------------------------
# 4. Fit multivariate trophic cascade model
# ------------------------------------------------------------

cascade_model <- brm(
  f_urchins +
    f_macrocystis +
    f_understory +
```

```
    f_abalone +
    set_rescor(FALSE),

  data = combined_wide_model,
  prior = priors,

  chains = 4,
  cores = 4,
  iter = 4000,
  warmup = 2000,

  control = list(adapt_delta = 0.95)
)
```

```
Compiling Stan program...


Trying to compile a simple C file


Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: ‘Apple clang version 17.0.0 (clang-1700.6.4.2)’
using SDK: ‘MacOSX26.2.sdk’
clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -
DNDEBUG   -I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/Rcpp/inclu
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/RcppEigen/include/"  -
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/RcppEigen/include/unsu
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/BH/include" -
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/StanHeaders/include/s
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/StanHeaders/include/" 
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/RcppParallel/include/
I"/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/rstan/include" -
DEIGEN_NO_DEBUG  -DBOOST_DISABLE_ASSERTS  -DBOOST_PENDING_INTEGER_LOG2_HPP  -
DSTAN_THREADS  -DUSE_STANC3 -DSTRICT_R_HEADERS  -DBOOST_PHOENIX_NO_VARIADIC_EXPRESSION  -
D_HAS_AUTO_PTR_ETC=0  -include '/Library/Frameworks/R.framework/Versions/4.3-
arm64/Resources/library/StanHeaders/include/stan/math/prim/fun/Eigen.hpp'  -
D_REENTRANT -DRCPP_PARALLEL_USE_TBB=1   -I/opt/R/arm64/include    -fPIC  -
falign-functions=64 -Wall -g -O2  -c foo.c -o foo.o
In file included from <built-in>:1:
In file included from /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/S
In file included from /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/R
In file included from /Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/R
/Library/Frameworks/R.framework/Versions/4.3-arm64/Resources/library/RcppEigen/include/Eigen/
```

```
  679 | #include <cmath>
      |          ^~~~~~~
1 error generated.
make: *** [foo.o] Error 1

Start sampling

Warning: There were 4 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems
```

```r
# ------------------------------------------------------------
# 5. Diagnostics
# ------------------------------------------------------------

summary(cascade_model)
```

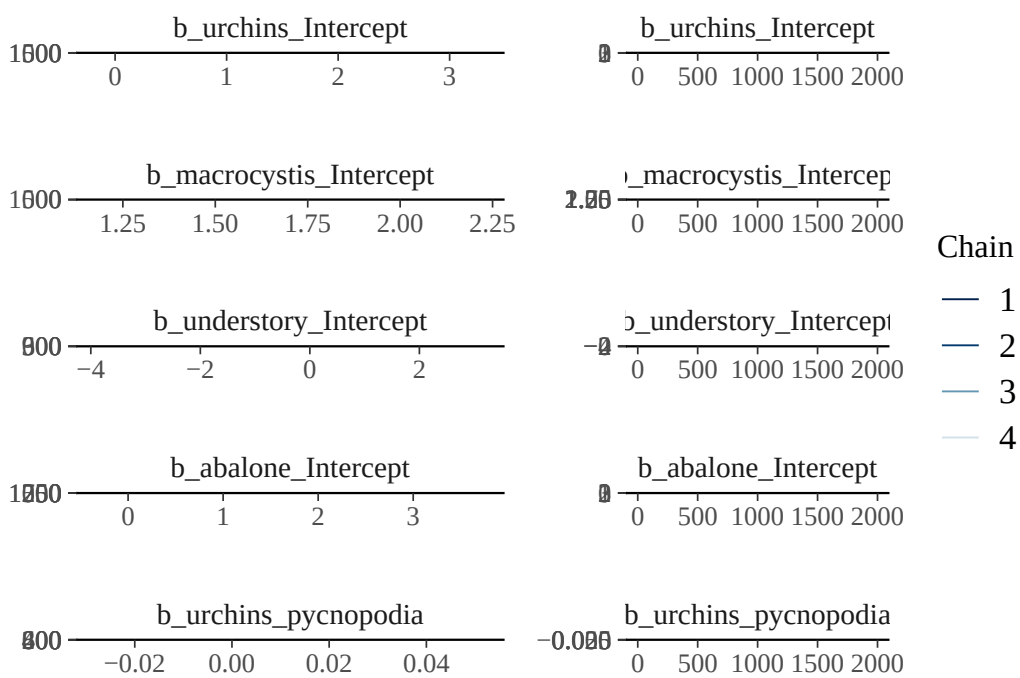
```
Warning: There were 4 divergent transitions after warmup. Increasing
adapt_delta above 0.95 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: MV(hurdle_gamma, hurdle_gamma, hurdle_gamma, hurdle_gamma)
  Links: mu = log; shape = identity; hu = identity
         mu = log; shape = identity; hu = identity
         mu = log; shape = identity; hu = identity
         mu = log; shape = identity; hu = identity
Formula: urchins ~ pycnopodia + (1 | year) + (1 | site/zone/transect/segment)
         macrocystis ~ urchins + (1 | year) + (1 | site/zone/transect/segment)
         understory ~ urchins + (1 | year) + (1 | site/zone/transect/segment)
         abalone ~ macrocystis + understory + pycnopodia + (1 | year) + (1 | site/zone/transe
   Data: combined_wide_model (Number of observations: 328)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~site (Number of levels: 5)
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(urchins_Intercept)         0.59      0.31     0.24     1.40 1.00     3487
```

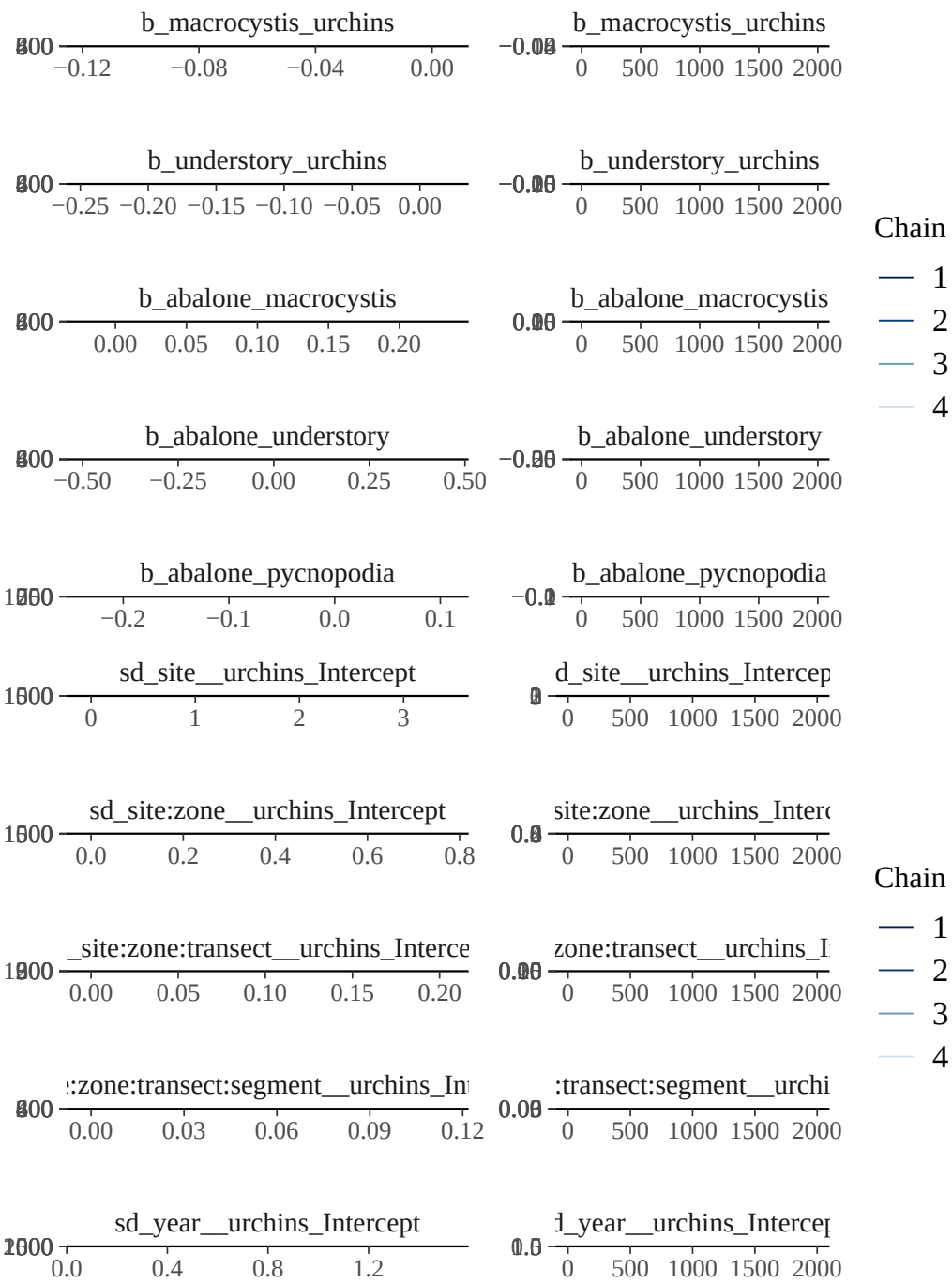
```
sd(macrocystis_Intercept)     0.07      0.08     0.00     0.24 1.00     4403
sd(understory_Intercept)      1.12      0.60     0.30     2.64 1.00     4388
sd(abalone_Intercept)         0.36      0.33     0.02     1.20 1.00     3628
                          Tail_ESS
sd(urchins_Intercept)         4387
sd(macrocystis_Intercept)     5421
sd(understory_Intercept)      3467
sd(abalone_Intercept)         4132

~site:zone (Number of levels: 10)
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(urchins_Intercept)         0.10      0.08     0.01     0.31 1.00     2017
sd(macrocystis_Intercept)     0.06      0.04     0.00     0.17 1.00     3771
sd(understory_Intercept)      0.50      0.26     0.16     1.16 1.00     3030
sd(abalone_Intercept)         0.28      0.19     0.01     0.73 1.00     2788
                          Tail_ESS
sd(urchins_Intercept)         3367
sd(macrocystis_Intercept)     4825
sd(understory_Intercept)      3951
sd(abalone_Intercept)         4518

~site:zone:transect (Number of levels: 20)
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(urchins_Intercept)         0.03      0.02     0.00     0.09 1.00     3621
sd(macrocystis_Intercept)     0.03      0.03     0.00     0.10 1.00     5565
sd(understory_Intercept)      0.11      0.09     0.00     0.32 1.00     3255
sd(abalone_Intercept)         0.38      0.12     0.17     0.65 1.00     3681
                          Tail_ESS
sd(urchins_Intercept)         4397
sd(macrocystis_Intercept)     4606
sd(understory_Intercept)      4407
sd(abalone_Intercept)         4837

~site:zone:transect:segment (Number of levels: 60)
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(urchins_Intercept)         0.02      0.02     0.00     0.07 1.00     4033
sd(macrocystis_Intercept)     0.03      0.02     0.00     0.09 1.00     5154
sd(understory_Intercept)      0.07      0.05     0.00     0.19 1.00     3857
sd(abalone_Intercept)         0.08      0.06     0.00     0.21 1.00     4721
                          Tail_ESS
sd(urchins_Intercept)         4171
sd(macrocystis_Intercept)     5078
sd(understory_Intercept)      4040
```

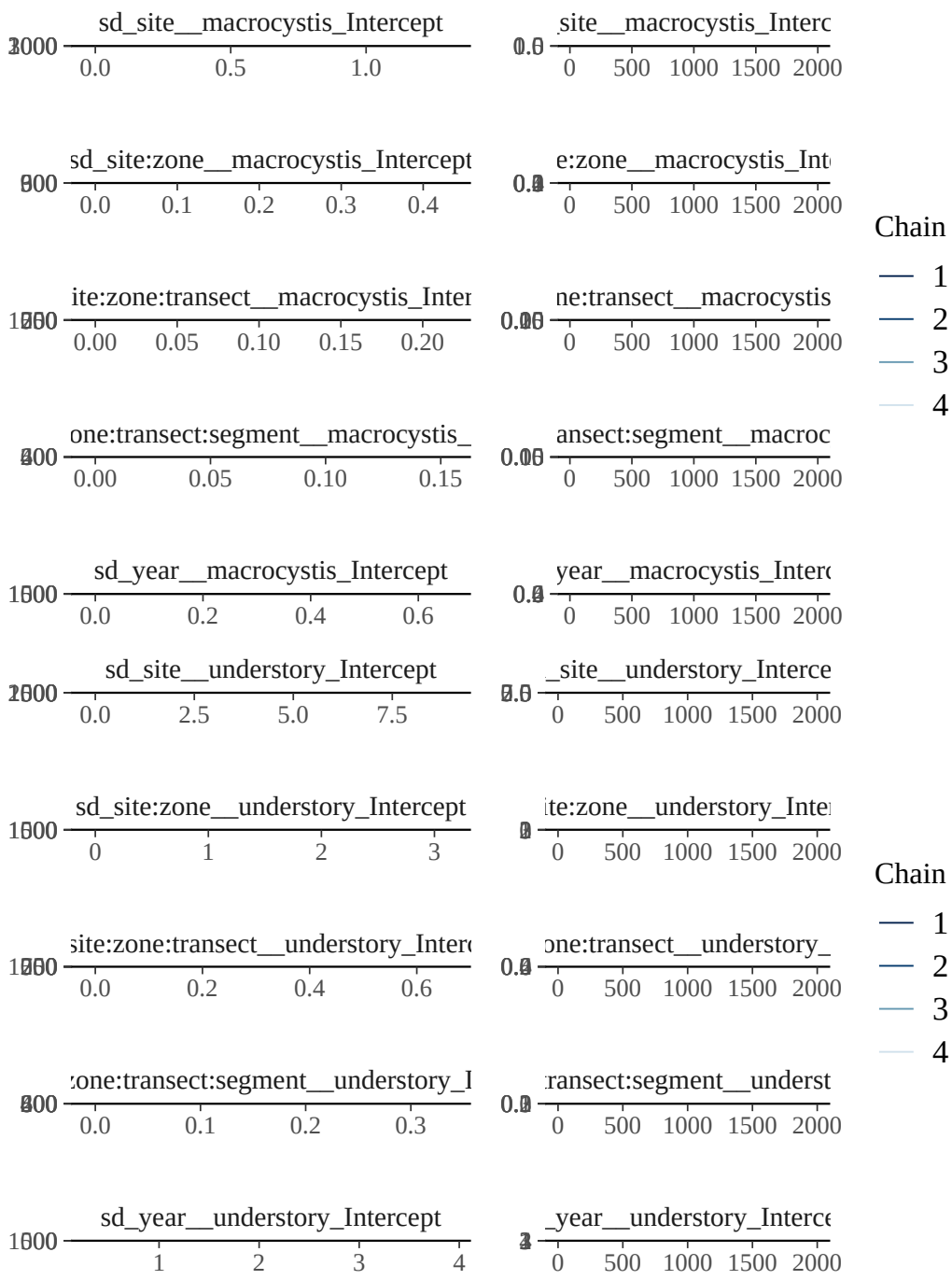
```
sd(abalone_Intercept)         4386

~year (Number of levels: 8)
                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
sd(urchins_Intercept)         0.25      0.09     0.14     0.49 1.00     2984
sd(macrocystis_Intercept)     0.09      0.06     0.01     0.25 1.00     2685
sd(understory_Intercept)      1.06      0.35     0.60     1.94 1.00     3783
sd(abalone_Intercept)         0.72      0.29     0.34     1.42 1.00     3099
                          Tail_ESS
sd(urchins_Intercept)         4766
sd(macrocystis_Intercept)     3936
sd(understory_Intercept)      4571
sd(abalone_Intercept)         4281

Regression Coefficients:
                      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
urchins_Intercept         1.41      0.30     0.83     2.03 1.00     3643
macrocystis_Intercept     1.75      0.09     1.58     1.92 1.00     8115
understory_Intercept     -0.30      0.68    -1.71     1.00 1.00     4165
abalone_Intercept         1.82      0.38     1.08     2.61 1.00     4527
urchins_pycnopodia        0.01      0.01    -0.01     0.03 1.00    14017
macrocystis_urchins      -0.05      0.02    -0.09    -0.02 1.00    12168
understory_urchins       -0.11      0.03    -0.18    -0.04 1.00    12524
abalone_macrocystis       0.12      0.03     0.06     0.18 1.00    11087
abalone_understory       -0.01      0.12    -0.24     0.22 1.00     9128
abalone_pycnopodia       -0.06      0.04    -0.13     0.01 1.00    11171
                      Tail_ESS
urchins_Intercept         4176
macrocystis_Intercept     5816
understory_Intercept      4810
abalone_Intercept         4942
urchins_pycnopodia        6237
macrocystis_urchins       5936
understory_urchins        6284
abalone_macrocystis       6041
abalone_understory        5793
abalone_pycnopodia        5731

Further Distributional Parameters:
                  Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
shape_urchins        17.22      1.45    14.50    20.18 1.00    11245     6316
shape_macrocystis    16.29      2.17    12.38    20.72 1.00    11302     5956
shape_understory      3.41      0.34     2.77     4.12 1.00    11069     5637
```

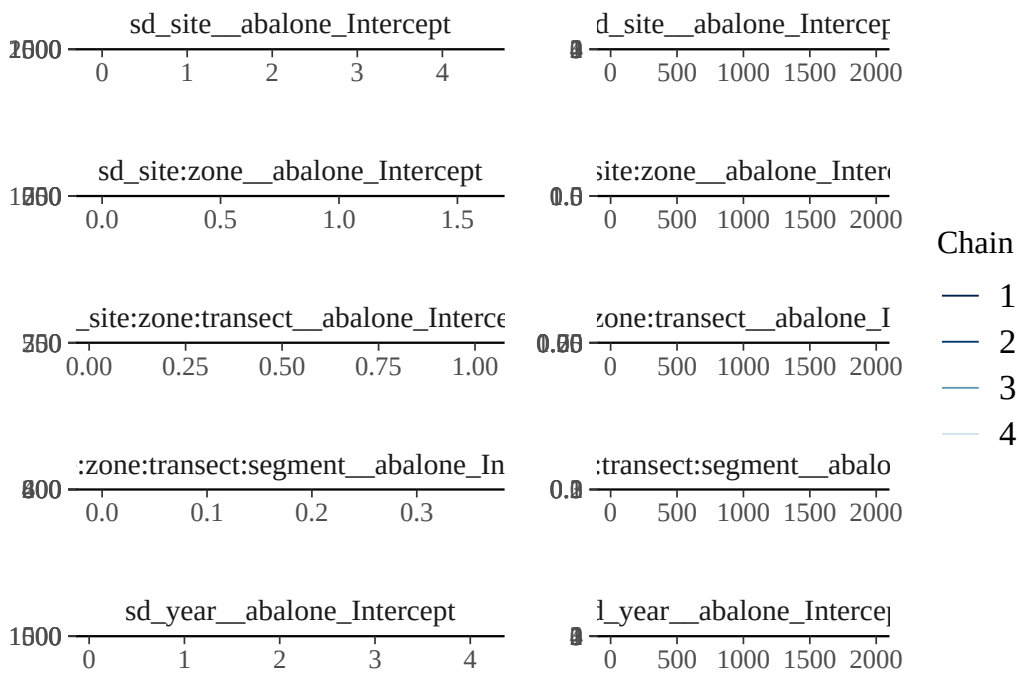
```
shape_abalone         1.42      0.12     1.20     1.67 1.00     9824     5975
hu_urchins            0.05      0.01     0.03     0.08 1.00    13140     4835
hu_macrocystis        0.61      0.03     0.56     0.66 1.00    13690     5816
hu_understory         0.38      0.03     0.33     0.44 1.00    15135     5508
hu_abalone            0.18      0.02     0.14     0.22 1.00    15177     6432

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(cascade_model)
```




b_macrocystis_urchins
b_macrocystis_urchins
b_understory_urchins
b_understory_urchins
b_abalone_macrocystis
b_abalone_macrocystis
b_abalone_understory
b_abalone_understory
b_abalone_pycnopodia
b_abalone_pycnopodia
sd_site__urchins_Intercept
d_site__urchins_Intercep
sd_site:zone__urchins_Intercept
site:zone__urchins_Interc
_site:zone:transect__urchins_Interce
zone:transect__urchins_I
:zone:transect:segment__urchins_In
:transect:segment__urchi
sd_year__urchins_Intercept
l_year__urchins_Intercep
Chain
1
2
3
4
Chain
1
2
3
4


sd_site__macrocystis_Intercept
site__macrocystis_Interc
sd_site:zone__macrocystis_Intercept
e:zone__macrocystis_Int
ite:zone:transect__macrocystis_Inter
ne:transect__macrocystis
one:transect:segment__macrocystis_
ansect:segment__macroc
sd_year__macrocystis_Intercept
year__macrocystis_Interc
sd_site__understory_Intercept
_site__understory_Interce
sd_site:zone__understory_Intercept
ite:zone__understory_Inte
site:zone:transect__understory_Interc
one:transect__understory_
zone:transect:segment__understory_I
ransect:segment__underst
sd_year__understory_Intercept
_year__understory_Interce
Chain
1
2
3
4
Chain
1
2
3
4


sd_site__abalone_Intercept
d_site__abalone_Intercep
sd_site:zone__abalone_Intercept
site:zone__abalone_Inter
_site:zone:transect__abalone_Interce
zone:transect__abalone_I
:zone:transect:segment__abalone_In
:transect:segment__abalo
sd_year__abalone_Intercept
l_year__abalone_Intercep
Chain
1
2
3
4